IMMERSIVE WATER INFRASTRUCTURE PLATFORM

WaterVision VR

Transforming **water infrastructure and collaboration** through Virtual Reality.



01 Project Overview

WHAT IT IS

- Collaborative **VR platform** for water engineers & operators
- Meet, inspect **3D models** & document decisions together
- Learn treatment processes through **immersive simulation**

02 Problem Statement

CHALLENGES

- Treatment processes are **hard to explain**
- **No immersive** training environments
- Weak **field–engineer** collaboration
- **Knowledge lost** after meetings
- Hard to **visualize** infrastructure

03 The Solution - Platform Modules

TEN CONNECTED CAPABILITIES

WaterVision VR Platform COLLABORATIVE CORE

01 Multi-user VR Meetings	02 Session Recording & Playback	03 Interactive Whiteboard	04 3D Model Inspection	05 Laser Pointer Collaboration
06 Screenshot Capture & Sharing	07 Wastewater Treatment Simulation	08 Sign Language Translation	09 AR Visualization Layer	10 Presence & Attendance Tracking

04 System Architecture

05 Main Features

DATA & SYNC FLOW

- VR Headsets - Meta Quest
- Unity Application
- Multiplayer Sync
- Simulation Engine
- Cloud Database & Analytics

DELIVERED ACROSS THREE SPRINTS

SPRINT 1

Foundation

- Multiplayer VR Environment
- Avatar Sync & Voice
- Rooms & Teleportation

SPRINT 2

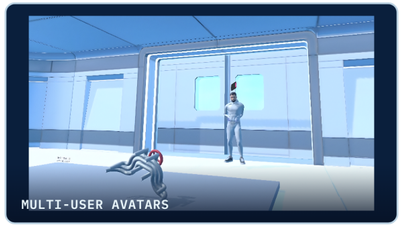
Collaboration

- Shared Whiteboard
- Interactive 3D Models
- Recording & Playback

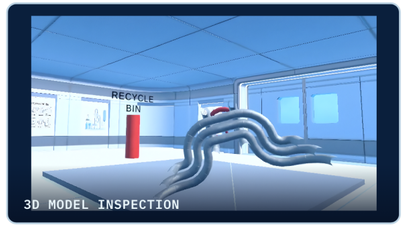
SPRINT 3

Advanced

- Augmented Reality (AR)
- Wastewater Simulation
- Sign Language Translation





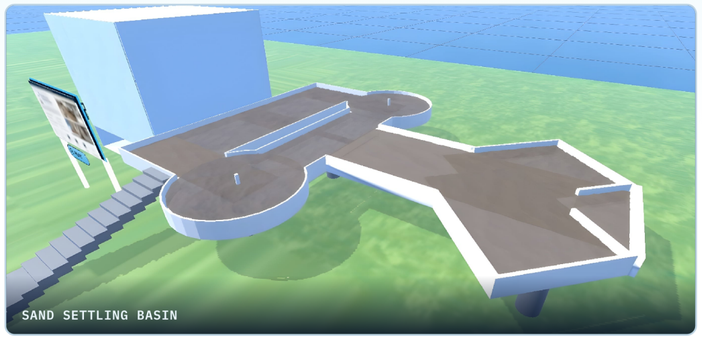


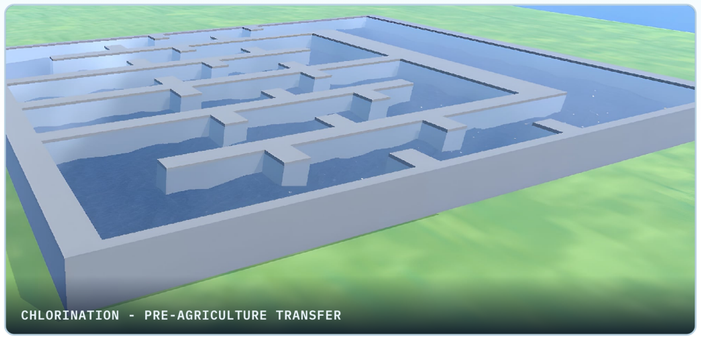


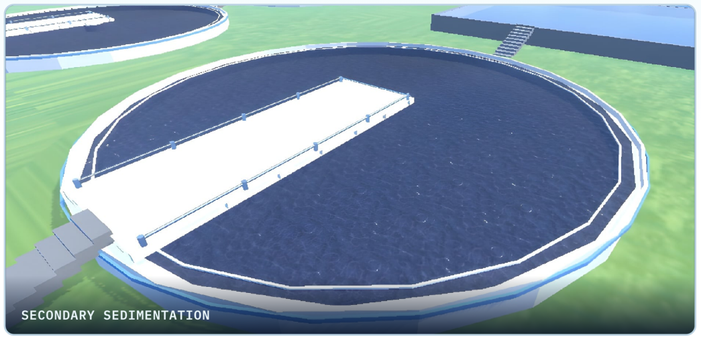
06 Wastewater Treatment Simulation

IMMERSIVE PROCESS FLOW

Wastewater Inlet (step 01) → Screening (step 02) → Primary Sedimentation (step 03) → Aeration Tank (step 04) → Secondary Clarifier (step 05) → Disinfection (step 06) → Clean Water Output (step 07)





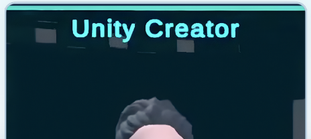


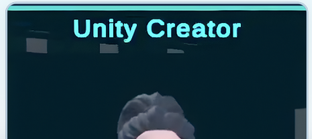
07 Sign Language Translation

INNOVATION · ACCESSIBILITY

Hand gestures are detected and translated into **sign language** inside VR.

- Accessible for **hearing-impaired** users
- **Inclusive** collaboration





08 Hand Tracking & Sync

HANDS-FREE INTERACTION



Better **learning** experience







- Hands tracked & **synced in real time**
- **Open the menu** by hand
- **Move & Teleport** - by hand

09 Methodology

AGILE SCRUM

1. **Sprint 1** — Foundation infrastructure
2. **Sprint 2** — Collaboration tools
3. **Sprint 3** — Advanced features

10 Results & Achievements

DELIVERED OUTCOMES

- Multi-user VR Collaboration
- Real-time Synchronization
- Session Documentation
- Interactive Learning Environment
- AR Integration
- Accessibility Features
- Water Treatment Simulation
- Professional Training Platform

11 Technologies

ENGINEERING STACK

- Unity
- C#
- VS Code
- Meta Quest
- XR Interaction Toolkit
- Netcode for GameObjects
- Virtual Reality
- Augmented Reality
- Agile Scrum
- Meshy AI
- Blender
- Claude

12 Project Team

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